

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

I Semester of BSc Biology Examination March 2018

BS101 Concepts of Physics - I

Date: 26-03-2018, Monday

Time: 10.00 AM To10.20 AM

Maximum Marks: 10

MCQ

[illegible]

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q-1 Choose the correct answer for the following questions

[10]

- _____ is a measure of the amount of deformation.
 - Hooke's law
 - stiffness of material
 - stress
 - strain
- The blood is a _____.
 - Newtonian fluid
 - ideal fluid
 - non-viscous fluid
 - non-Newtonian fluid
- The second law of thermodynamics states that the total entropy of a closed system _____.
 - always decreases
 - always increases
 - always zero
 - always changes
- For _____, the regime of Reynolds number is $0 < R < 2000$.
 - turbulent flow
 - laminar flow
 - unsteady flow
 - time-dependent flow
- If the net force is 100 N at 90° angle, the component of force (F_x) in X-direction is _____.
 - 100 N
 - 50 N
 - 0 N
 - 50 N
- _____ is the equation of state of an ideal gas.
 - $PV = nRT$
 - $PV = R$
 - $P_1V_1 = P_2V_2$
 - $P_1/T_1 = P_2/T_2$

7. The pitch is defined as the characteristic of sound by which a particular sound appears to be _____.

a) loud	b) feeble
c) grave or shrill	d) high intensity
8. Full form of AFM is _____.

a) Atomic Force Microscopy	b) Automatic Force Microscopy
c) Atomic Force Microscope	d) Atomic Force Measurement
9. _____ is the distance between two identical adjacent points in a wave.

a) Wave number	b) Wavelength
c) Frequency	d) Time period
10. The freezing point of water is _____.

a) 273 K	b) -273 K
c) 300 K	d) -300 K

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I Semester of BSc Biology Examination March 2018

BS101 Concepts of Physics – I

Date: 26-03-2018, Monday

Time: 10.20 AM To 12.00 PM

Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I

Q - II Answer the following questions as directed **10**

1. Define loudness and intensity. 2
2. Explain Work–Energy Theorem in brief. 2
3. Discuss zeroth law thermodynamics with proper figure. 2
4. Discuss surface tension with proper equation. 2
5. Describe the superposition principle. 2
6. Write the ideal gas law and explain each term. 2
7. Boat displaces 5 m^3 of water and density of water is 1000 kg/m^3 . What is the mass of water displaced? What is the buoyant force? 2

SECTION - II

Q-III Answer the following questions as directed **15**

1. Define Doppler effect and discuss any two cases of it. 3

OR

A spring stretches by 3.9 cm when a 10 gm mass is hung from it. If a 25 gm mass attached to this spring oscillates in simple harmonic motion, calculate the period of motion. 3

2. Estimate Young's modulus for compression of bone from the following data. The femur of 80 kg person has an effective cross-sectional area of about 4 cm^2 and a length of about 0.4 m. When the person lifts a 40 kg mass, careful measurements show that the femur compresses by about 0.03 mm. 3

3. Prove work-energy relation using translational kinematics. 3

OR

Explain linear thermal expansion and find the thermal stress using the young modulus. 3

4. Define entropy and discuss the second law of thermodynamics with proper equation. 3

5. Derive the expression for the conservation of energy (Bernoulli's Equation) for an ideal fluid. 3

OR

What is ultrasound? Write two properties and two applications. 3